

Microsoft open sources 'pg_durable' extension

Microsoft has open sourced `pg_durable`, a PostgreSQL extension that lets developers build long-running, fault-tolerant software workflows directly inside the database.

The extension introduces durable execution natively to PostgreSQL, eliminating the need for developers to patch together external tools like background workers, message queues, and external orchestration engines. Workflows are defined using standard SQL, allowing PostgreSQL to automatically track execution progress, handle retries, and recover states following unexpected server crashes or database restarts.

By shifting workflow state tracking directly into PostgreSQL tables, `pg_durable` guarantees that long-running data processes can survive unexpected system failures and automatically pick up from their last recorded checkpoint. The extension provides developers with dedicated SQL operators designed specifically for managing sequential steps and parallel tasks, making it highly optimised for asynchronous tasks.

Architecturally minimal, the open source extension runs via a native Postgres background worker with no external control plane requirements. The underlying orchestration runtime is powered by two new Microsoft-developed Rust libraries: `durooxide`, which coordinates deterministic event tracking, and `durooxide-pg`, which handles state persistence in a dedicated schema.

The `pg_durable` extension is available on GitHub and ships natively as part of Microsoft's Azure HorizonDB cloud service.



The model is built on Qwen3.6-27B with the base parameters remaining frozen. VeriLoop Coder-E1 employs narrow-domain parameter-efficient fine-tuning (PEFT) delivered via a detachable surface host adaptor to reinforce tool-contract following, evidence-conclusion binding, uncertainty identification, failure explanation, local repair, and rollback control.

The system utilises a proprietary self-harness mechanism that structures code generation as a closed, evidence-governed multi-round loop, recompiling generated code, test receipts, tool feedback, and counter-evidence into structured Markdown.

In benchmark evaluations, VeriLoop Coder-E1 demonstrated state-of-the-art capabilities for open source vertical coding models, achieving impressive scores of 85.20 on SWE-bench Verified, 62.38 on SWE-bench Pro, 76.40 on Terminal-Bench 2.0, and 33.63 on DeepSWE.

Among open source models at or under 32B parameters, VeriLoop Coder-E1 ranks first on SWE-bench Verified, SWE-bench Pro, and Terminal-Bench 2.0, and second on DeepSWE. The core model weights and adaptor assets are fully open sourced, whilst the upper-level self-harness control stack remains retained by the team.

Huawei's openPangu pretrains on Ascend NPUs without using Nvidia GPUs

Huawei has released the weights, inference code and technical report for openPangu-2.0-Pro, a 505-billion-parameter open-weight language model that it says completed its entire pretraining run on Ascend 910B NPUs without using Nvidia GPUs. If validated, it would be the first publicly released open-weight model above 500 billion parameters to achieve that milestone.

The model is now available through GitCode's Ascend Tribe community and Huawei Cloud ModelArts Studio, allowing developers to download the weights immediately. Community efforts to convert the model for Nvidia GPU inference are also underway, broadening access beyond Huawei's hardware ecosystem.

Built on a Mixture of Experts (MoE) architecture with 18 billion active parameters per token, openPangu-2.0-Pro features a 512,000-token context window and was pretrained on approximately 34 trillion tokens.

However, the release does not establish a fully domestic Chinese AI hardware supply chain.

Bangladesh university develops 32-bit RISC-V processor with open source design tools

A research team at the Bangladesh University of Engineering and Technology (BUET) has designed a 32-bit processor using only open source technologies. Built on the RISC-V instruction set architecture (ISA), the processor was developed without proprietary electronic design automation (EDA) software, highlighting how freely available tools can support advanced semiconductor research while reducing development costs.

The processor was designed as a complete system on a chip (SoC) in which the design, verification and tape-out were all carried out using free EDA software and process design kits. The resulting chip was manufactured by IHP – Leibniz Institute for High Performance Microelectronics of Germany using their open source manufacturing services.

Developed by undergraduate students in cooperation with faculty members, the processor is meant to be used in embedded applications, including appliances and IoT products.

This example demonstrates how open hardware communities can help educational institutions design their own processor solutions without spending money on expensive proprietary systems.